

C41

1. (a)

$$\begin{aligned} \int |\psi_{\pm}|^2 dr_1 dr_2 &= |A|^2 \int [\psi_a(r_1) \psi_b(r_2) \pm \psi_b(r_1) \psi_a(r_2)]^* [\psi_a(r_1) \psi_b(r_2) \\ &\quad \pm \psi_b(r_1) \psi_a(r_2)] dr_1 dr_2 \\ &= |A|^2 \left[\int |\psi_a(r_1)|^2 dr_1 \int |\psi_b(r_2)|^2 dr_2 \pm \int \psi_b^*(r_1) \psi_a(r_1) dr_1 \int \psi_a^*(r_2) \psi_b(r_2) dr_2 \right. \\ &\quad \left. \pm \int \psi_a^*(r_1) \psi_b(r_1) dr_1 \int \psi_b^*(r_2) \psi_a(r_2) dr_2 + \int |\psi_b(r_1)|^2 dr_1 \int |\psi_a(r_2)|^2 dr_2 \right] \\ &= |A|^2 (1 \pm 0 \pm 0 + 1) \\ &= 2|A|^2 = 1 \end{aligned}$$

$$\text{Then } A = \frac{1}{\sqrt{2}}$$

(b) $\psi_a = \psi_b$

$$\psi_{+}(r_1, r_2) = 2A \psi_a(r_1) \psi_b(r_2)$$

$$\begin{aligned} \int |\psi_{+}(r_1, r_2)|^2 dr_1 dr_2 &= |A|^2 \int [2\psi_a(r_1) \psi_b(r_2)]^* [2\psi_a(r_1) \psi_b(r_2)] dr_1 dr_2 \\ &= 4|A|^2 \int |\psi_a(r_1)|^2 dr_1 \int |\psi_b(r_2)|^2 dr_2 \\ &= 4|A|^2 = 1 \end{aligned}$$

$$\text{Then } A = \frac{1}{2}$$

$$2.(a) E = (n_1 + n_2 + n_3 + \frac{3}{2}) \hbar \omega = \frac{9}{2} \hbar \omega, \text{ 则 } n_1 + n_2 + n_3 = 3$$

可能的排列方式:

$$(1, 1, 1)$$

$$(0, 0, 3) (0, 3, 0) (3, 0, 0)$$

$$(0, 1, 2) (0, 2, 1) (1, 0, 2) (1, 2, 0) (2, 0, 1) (2, 1, 0)$$

Configuration 1: $(0, 3, 0, 0, 0, \dots)$, 1 distinct state, $Q = \frac{1}{10}$

Configuration 2: $(2, 0, 0, 1, 0, \dots)$, 3 distinct state, $Q = \frac{3}{10}$

Configuration 3: $(1, 1, 1, 0, 0, \dots)$, 6 distinct state, $Q = \frac{6}{10}$

$\therefore$ Configuration 3 is the most probable configuration.

$$\text{for } E_0 = \frac{1}{2} \hbar \omega, P_0 = \frac{3}{10} \times \frac{2}{3} + \frac{6}{10} \times \frac{1}{3} = \frac{4}{10} = \frac{2}{5}$$

$$\text{for } E_1 = \frac{3}{2} \hbar \omega, P_1 = \frac{1}{10} \times 1 + \frac{6}{10} \times \frac{1}{3} = \frac{3}{10}$$

$$\text{for } E_2 = \frac{5}{2} \hbar \omega, P_2 = \frac{6}{10} \times \frac{1}{3} = \frac{2}{10} = \frac{1}{5}$$

$$\text{for } E_3 = \frac{7}{2} \hbar \omega, P_3 = \frac{3}{10} \times \frac{1}{3} = \frac{1}{10}$$

$\therefore E_0$ is the most probable energy. probability: $\frac{2}{5}$

(b) 若费米子, 则泡利不相容原理, 只有 $(1, 1, 1, 0, 0, \dots)$ 允许.

$\therefore$ this is the most probable configuration.

$$\text{for } E_0 = \frac{1}{2} \hbar \omega, P_0 = \frac{1}{3}, \text{ for } E_1 = \frac{3}{2} \hbar \omega, P_1 = \frac{1}{3}$$

$$\text{for } E_2 = \frac{5}{2} \hbar \omega, P_2 = \frac{1}{3}$$

三个能级的能量对应的概率相同. 故均为最概能能量.

(c) 若波色子, 则不可区分.

Configuration 1: $(0, 3, 0, 0, 0, \dots)$, 1 distinct state, $Q = \frac{1}{3}$

Configuration 2: $(2, 0, 0, 1, 0, \dots)$, 1 distinct state, $Q = \frac{1}{3}$

Configuration 3: $(1, 1, 1, 0, 0, \dots)$, 1 distinct state, $Q = \frac{1}{3}$

for $E_0 = \frac{1}{2}\hbar\omega$, $P_0 = \frac{1}{3}$, for $E_1 = \frac{3}{2}\hbar\omega$, $P_1 = \frac{1}{3} \times 1 + \frac{1}{3} \times \frac{1}{3} = \frac{4}{9}$

for $E_2 = \frac{5}{2}\hbar\omega$, $P_2 = \frac{1}{3} \times \frac{1}{3} = \frac{1}{9}$, for $E_3 = \frac{7}{2}\hbar\omega$, $P_3 = \frac{1}{3} \times \frac{1}{3} = \frac{1}{9}$

$\therefore E_1 = \frac{3}{2}\hbar\omega$ 为最可能能量, $P_1 = \frac{4}{9}$

$$3. P = C_{10}^4 \cdot C_6^3 \cdot C_3^2 \cdot C_1^1$$

$$= 210 \times 20 \times 3 \times 1$$

$$= 12600$$